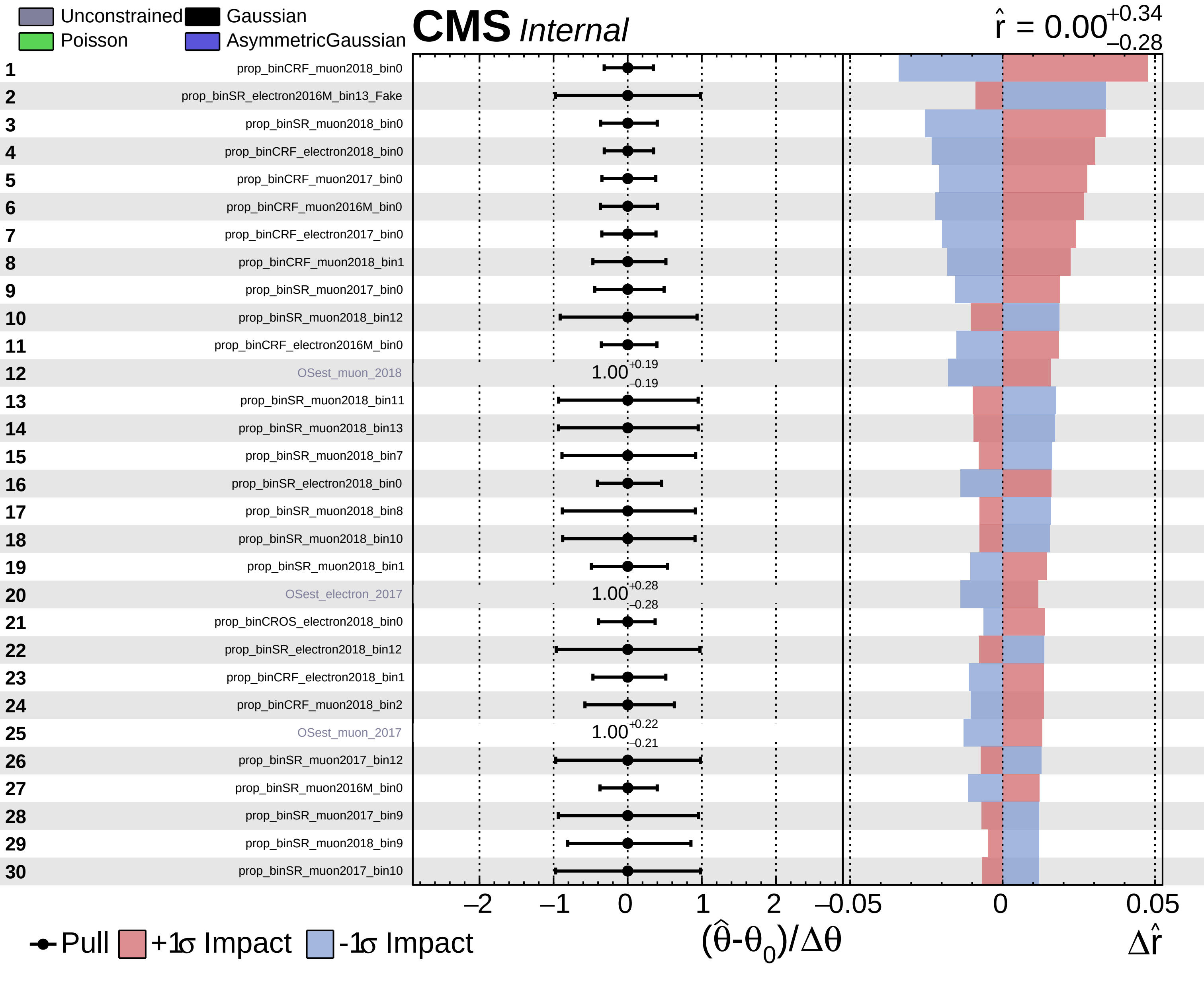
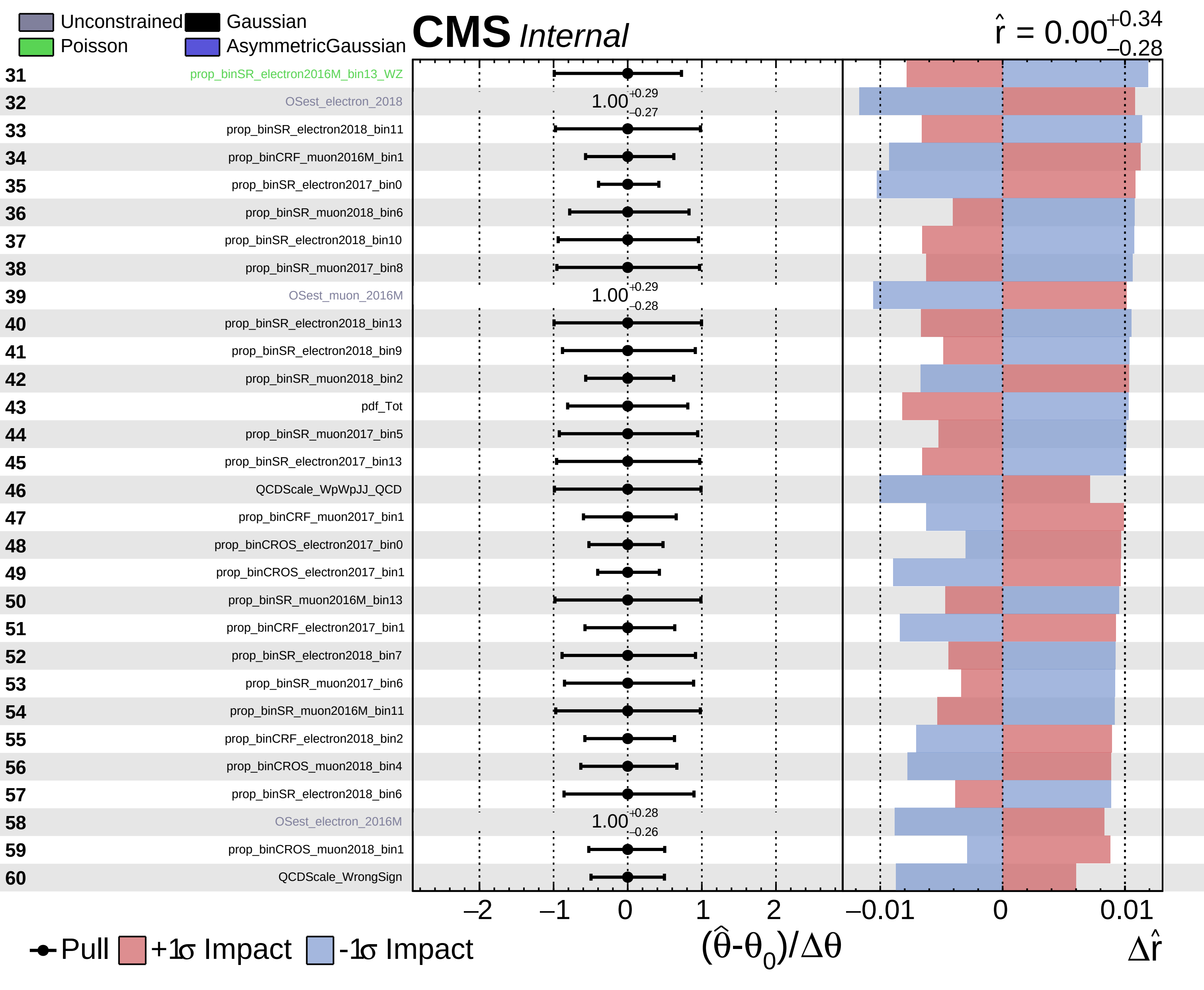
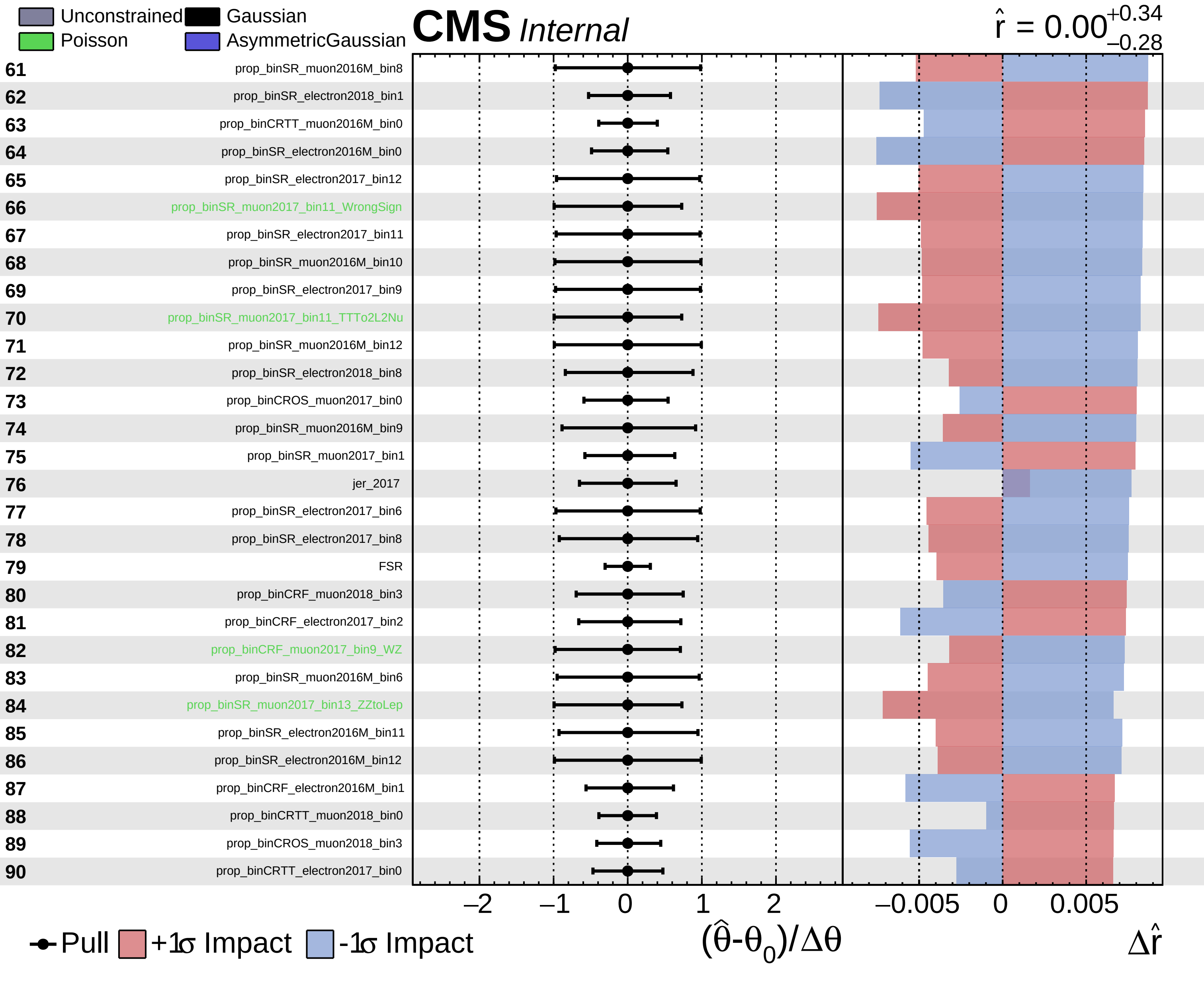
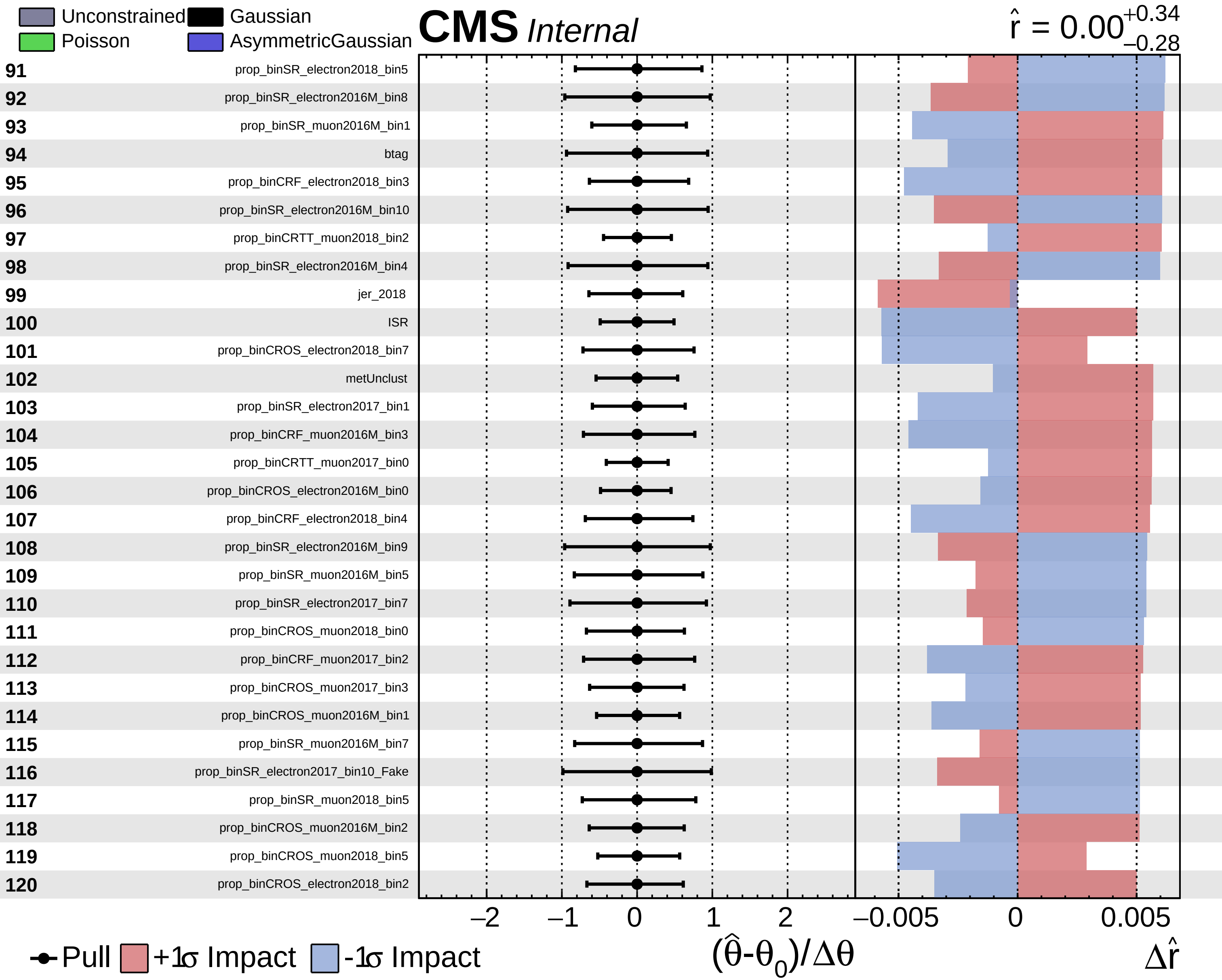

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
1 prop_binCRF_muon2018_bin0
2 prop_binSR_electron2016M_bin13_Fake
3 prop_binSR_muon2018_bin0
4 prop_binCRF_electron2018_bin0
5 prop_binCRF_muon2017_bin0
6 prop_binCRF_muon2016M_bin0
7 prop_binCRF_electron2017_bin0
8 prop_binCRF_muon2018_bin1
9 prop_binSR_muon2017_bin0
10 prop_binSR_muon2018_bin12
11 prop_binCRF_electron2016M_bin0
12 OSest_muon_2018
$1.00^{+0.19}_{-0.19}$
13 prop_binSR_muon2018_bin11
14 prop_binSR_muon2018_bin13
15 prop_binSR_muon2018_bin7
16 prop_binSR_electron2018_bin0
17 prop_binSR_muon2018_bin8
18 prop_binSR_muon2018_bin10
19 prop_binSR_muon2018_bin1
20 OSest_electron_2017
$1.00^{+0.28}_{-0.28}$
21 prop_binCROS_electron2018_bin0
22 prop_binSR_electron2018_bin12
23 prop_binCRF_electron2018_bin1
24 prop_binCRF_muon2018_bin2
25 OSest_muon_2017
$1.00^{+0.22}_{-0.21}$
26 prop_binSR_muon2017_bin12
27 prop_binSR_muon2016M_bin0
28 prop_binSR_muon2017_bin9
29 prop_binSR_muon2018_bin9
30 prop_binSR_muon2017_bin10
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

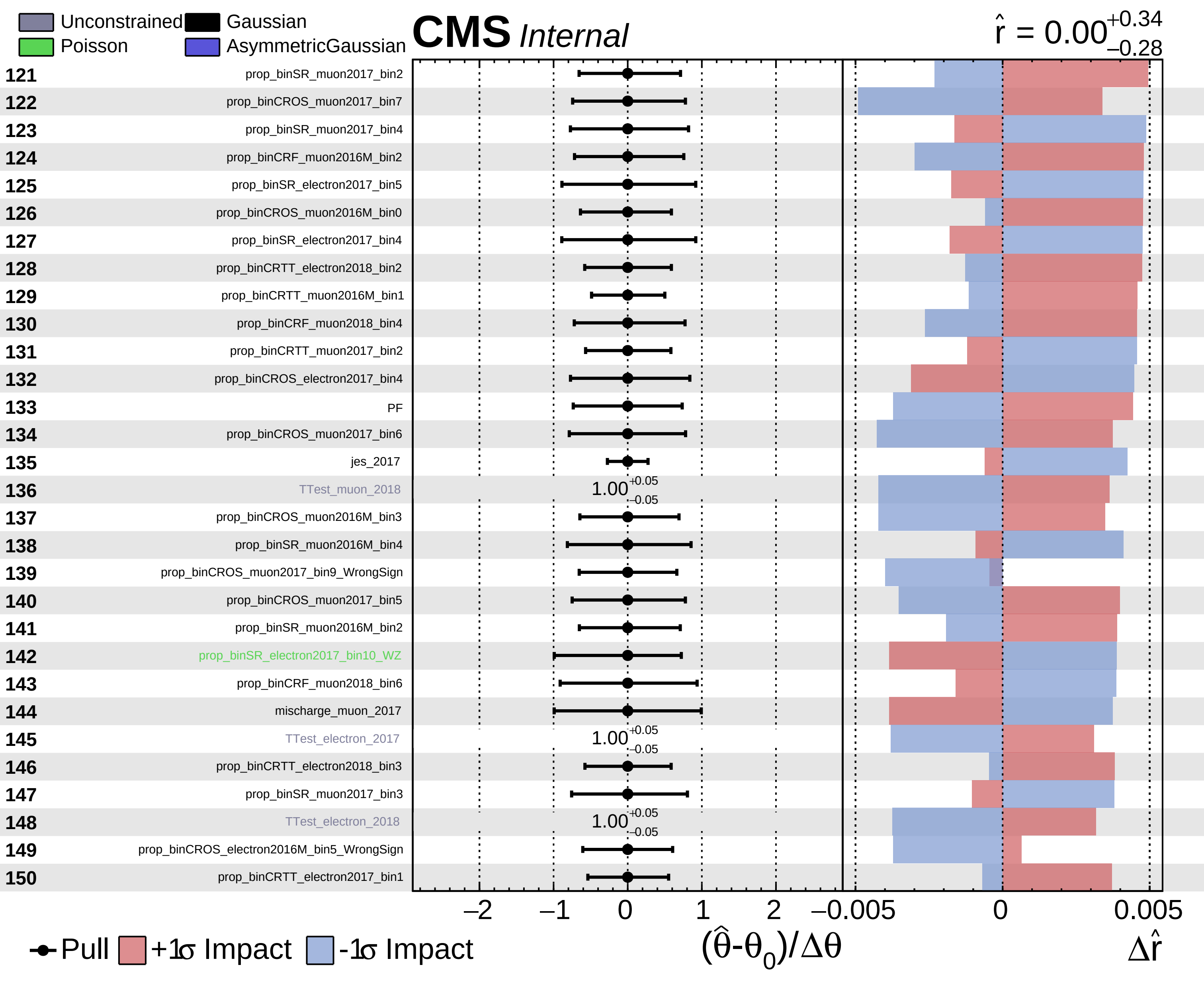


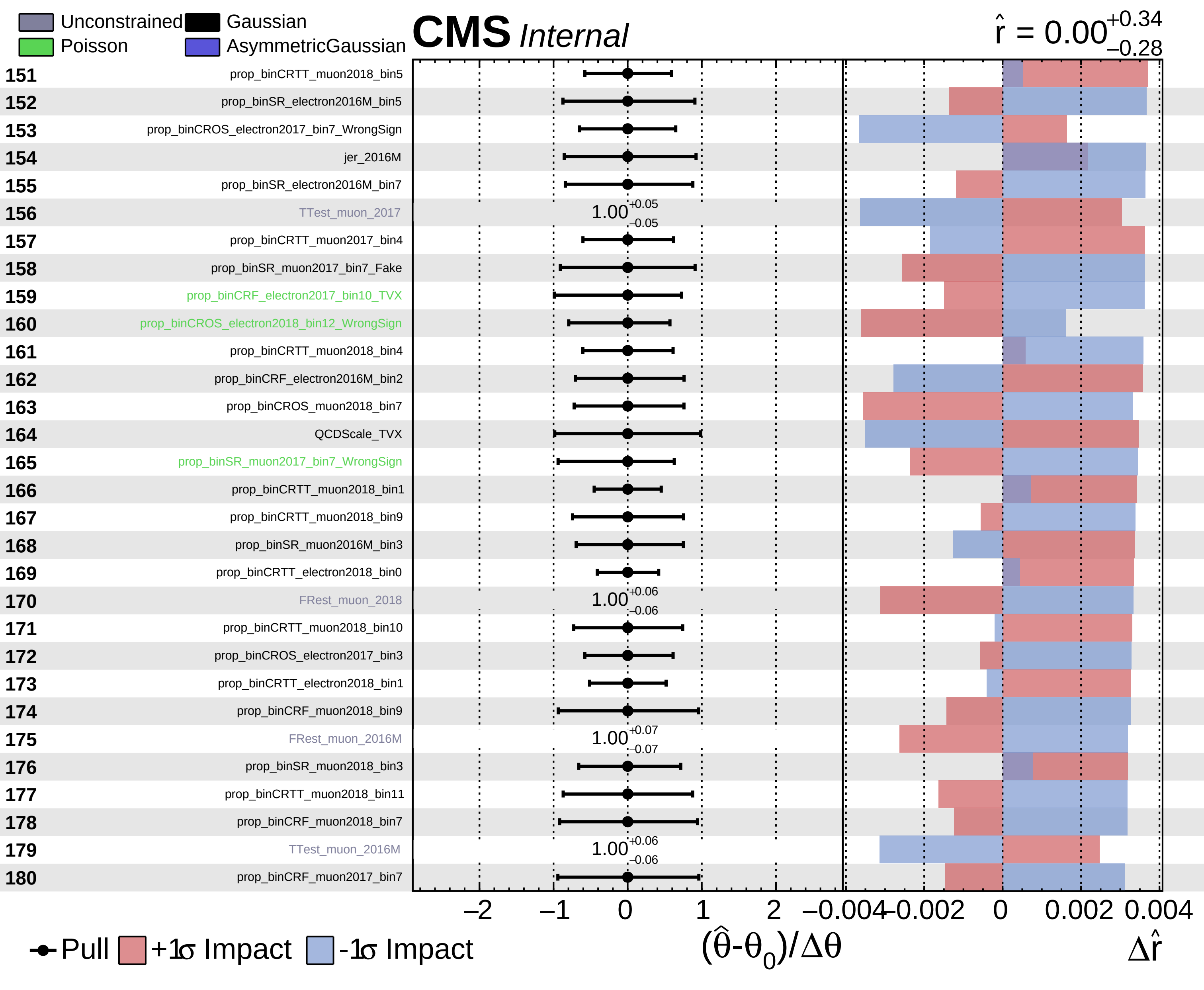
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
31 prop_binSR_electron2016M_bin13_WZ
32 OSest_electron_2018 $1.00^{+0.29}_{-0.27}$
33 prop_binSR_electron2018_bin11
34 prop_binCRF_muon2016M_bin1
35 prop_binSR_electron2017_bin0
36 prop_binSR_muon2018_bin6
37 prop_binSR_electron2018_bin10
38 prop_binSR_muon2017_bin8
39 OSest_muon_2016M $1.00^{+0.29}_{-0.28}$
40 prop_binSR_electron2018_bin13
41 prop_binSR_electron2018_bin9
42 prop_binSR_muon2018_bin2
43 pdf_Tot
44 prop_binSR_muon2017_bin5
45 prop_binSR_electron2017_bin13
46 QCDScale_WpWpJJ_QCD
47 prop_binCRF_muon2017_bin1
48 prop_binCROS_electron2017_bin0
49 prop_binCROS_electron2017_bin1
50 prop_binSR_muon2016M_bin13
51 prop_binCRF_electron2017_bin1
52 prop_binSR_electron2018_bin7
53 prop_binSR_muon2017_bin6
54 prop_binSR_muon2016M_bin11
55 prop_binCRF_electron2018_bin2
56 prop_binCROS_muon2018_bin4
57 prop_binSR_electron2018_bin6
58 OSest_electron_2016M $1.00^{+0.28}_{-0.26}$
59 prop_binCROS_muon2018_bin1
60 QCDScale_WrongSign
−2 −1 0 1 2
−0.01 0 0.01
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
61 prop_binSR_muon2016M_bin8
62 prop_binSR_electron2018_bin1
63 prop_binCRTT_muon2016M_bin0
64 prop_binSR_electron2016M_bin0
65 prop_binSR_electron2017_bin12
66 prop_binSR_muon2017_bin11_WrongSign
67 prop_binSR_electron2017_bin11
68 prop_binSR_muon2016M_bin10
69 prop_binSR_electron2017_bin9
70 prop_binSR_muon2017_bin11_TTTo2L2Nu
71 prop_binSR_muon2016M_bin12
72 prop_binSR_electron2018_bin8
73 prop_binCROS_muon2017_bin0
74 prop_binSR_muon2016M_bin9
75 prop_binSR_muon2017_bin1
76 jer_2017
77 prop_binSR_electron2017_bin6
78 prop_binSR_electron2017_bin8
79 FSR
80 prop_binCRF_muon2018_bin3
81 prop_binCRF_electron2017_bin2
82 prop_binCRF_muon2017_bin9_WZ
83 prop_binSR_muon2016M_bin6
84 prop_binSR_muon2017_bin13_ZZtoLep
85 prop_binSR_electron2016M_bin11
86 prop_binSR_electron2016M_bin12
87 prop_binCRF_electron2016M_bin1
88 prop_binCRTT_muon2018_bin0
89 prop_binCROS_muon2018_bin3
90 prop_binCRTT_electron2017_bin0
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

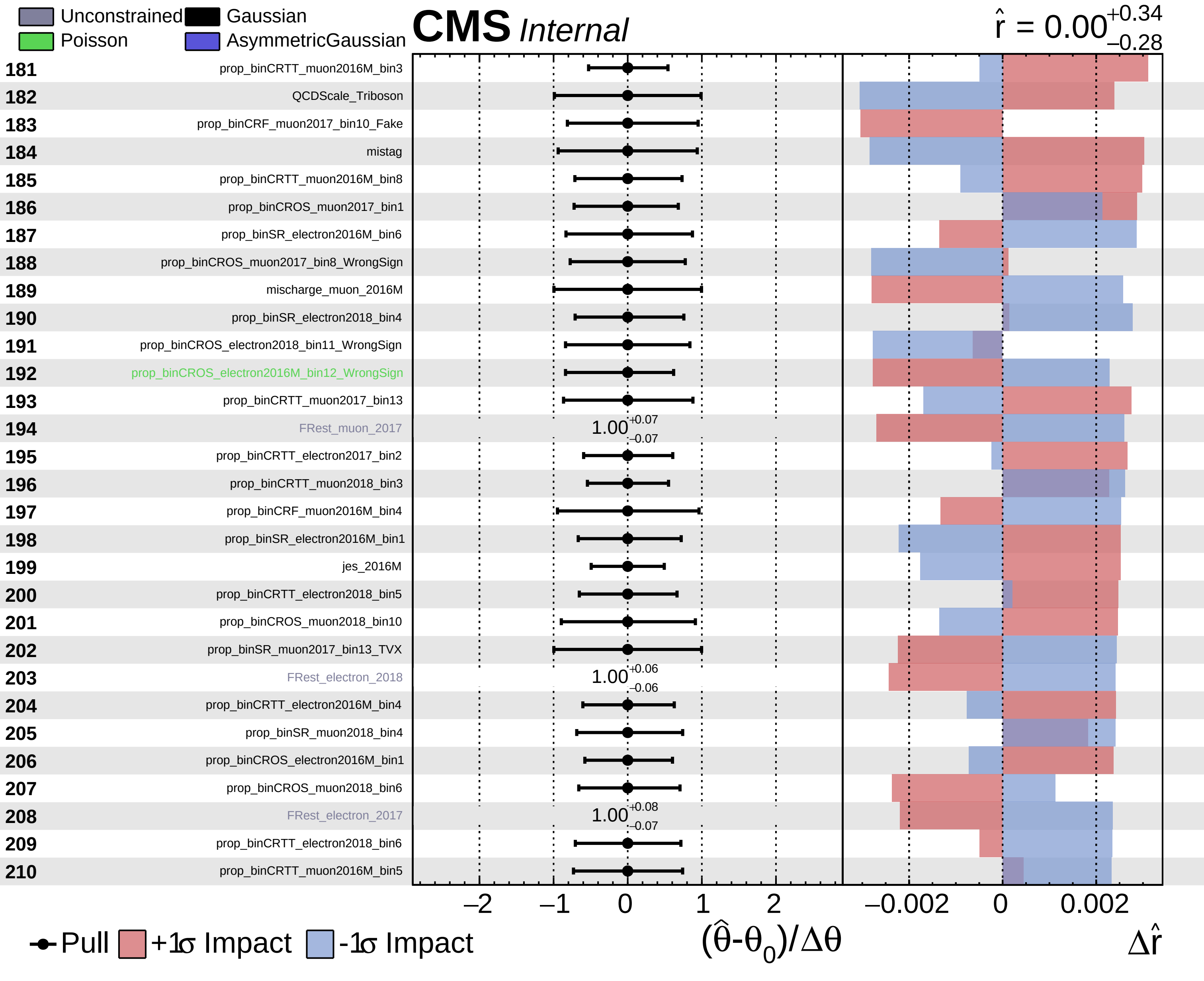


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
91 prop_binSR_electron2018_bin5
92 prop_binSR_electron2016M_bin8
93 prop_binSR_muon2016M_bin1
94 btag
95 prop_binCRF_electron2018_bin3
96 prop_binSR_electron2016M_bin10
97 prop_binCRTT_muon2018_bin2
98 prop_binSR_electron2016M_bin4
99 jer_2018
100 ISR
101 prop_binCROS_electron2018_bin7
102 metUnclust
103 prop_binSR_electron2017_bin1
104 prop_binCRF_muon2016M_bin3
105 prop_binCRTT_muon2017_bin0
106 prop_binCROS_electron2016M_bin0
107 prop_binCRF_electron2018_bin4
108 prop_binSR_electron2016M_bin9
109 prop_binSR_muon2016M_bin5
110 prop_binSR_electron2017_bin7
111 prop_binCROS_muon2018_bin0
112 prop_binCRF_muon2017_bin2
113 prop_binCROS_muon2017_bin3
114 prop_binCROS_muon2016M_bin1
115 prop_binSR_muon2016M_bin7
116 prop_binSR_electron2017_bin10_Fake
117 prop_binSR_muon2018_bin5
118 prop_binCROS_muon2016M_bin2
119 prop_binCROS_muon2018_bin5
120 prop_binCROS_electron2018_bin2
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

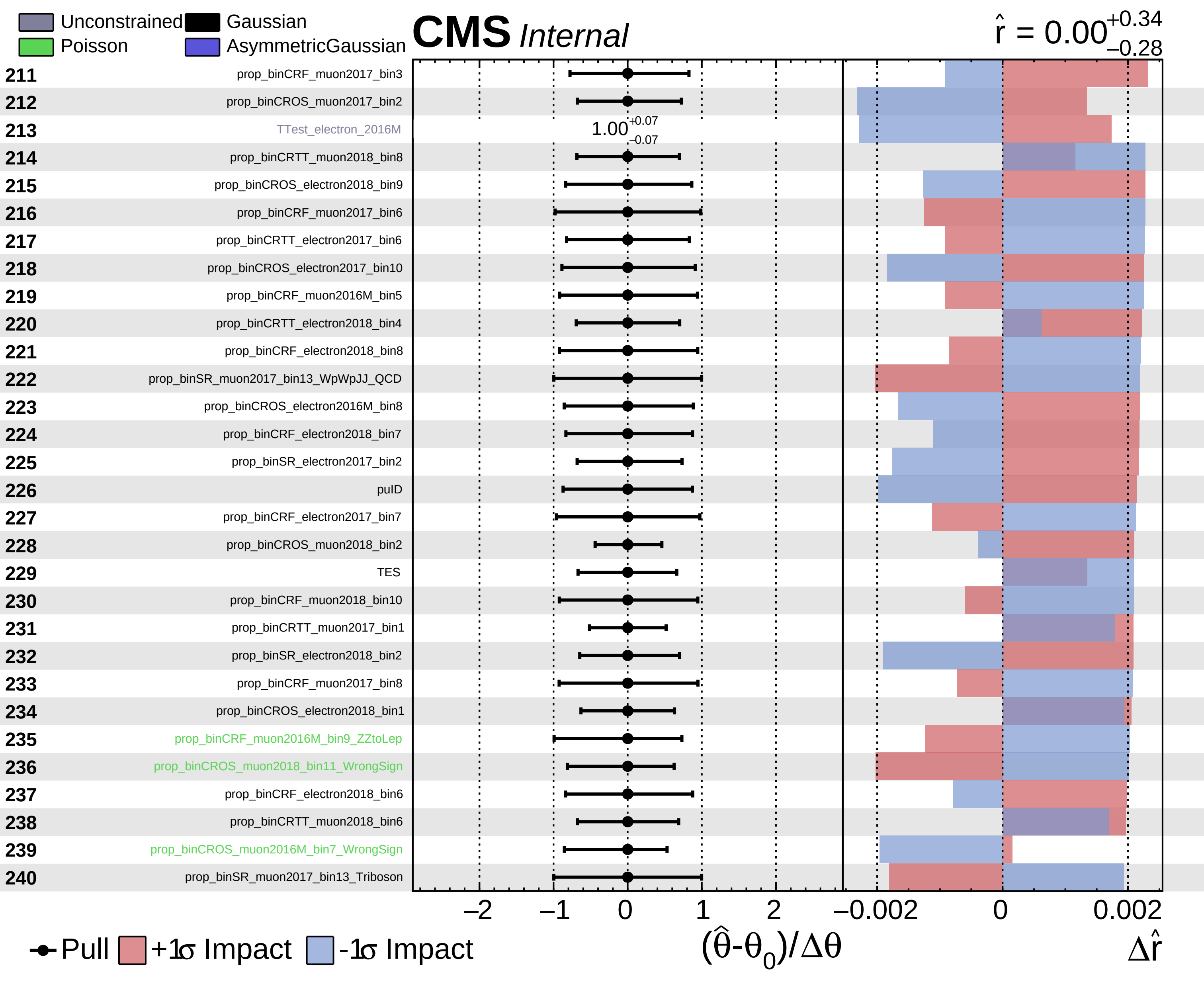

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
121 prop_binSR_muon2017_bin2
122 prop_binCROS_muon2017_bin7
123 prop_binSR_muon2017_bin4
124 prop_binCRF_muon2016M_bin2
125 prop_binSR_electron2017_bin5
126 prop_binCROS_muon2016M_bin0
127 prop_binSR_electron2017_bin4
128 prop_binCRTT_electron2018_bin2
129 prop_binCRTT_muon2016M_bin1
130 prop_binCRF_muon2018_bin4
131 prop_binCRTT_muon2017_bin2
132 prop_binCROS_electron2017_bin4
133 PF
134 prop_binCROS_muon2017_bin6
135 jes_2017
136 TTest_muon_2018
$1.00^{+0.05}_{-0.05}$
137 prop_binCROS_muon2016M_bin3
138 prop_binSR_muon2016M_bin4
139 prop_binCROS_muon2017_bin9_WrongSign
140 prop_binCROS_muon2017_bin5
141 prop_binSR_muon2016M_bin2
142 prop_binSR_electron2017_bin10_WZ
143 prop_binCRF_muon2018_bin6
144 mischarge_muon_2017
145 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
146 prop_binCRTT_electron2018_bin3
147 prop_binSR_muon2017_bin3
148 TTest_electron_2018
$1.00^{+0.05}_{-0.05}$
149 prop_binCROS_electron2016M_bin5_WrongSign
150 prop_binCRTT_electron2017_bin1
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



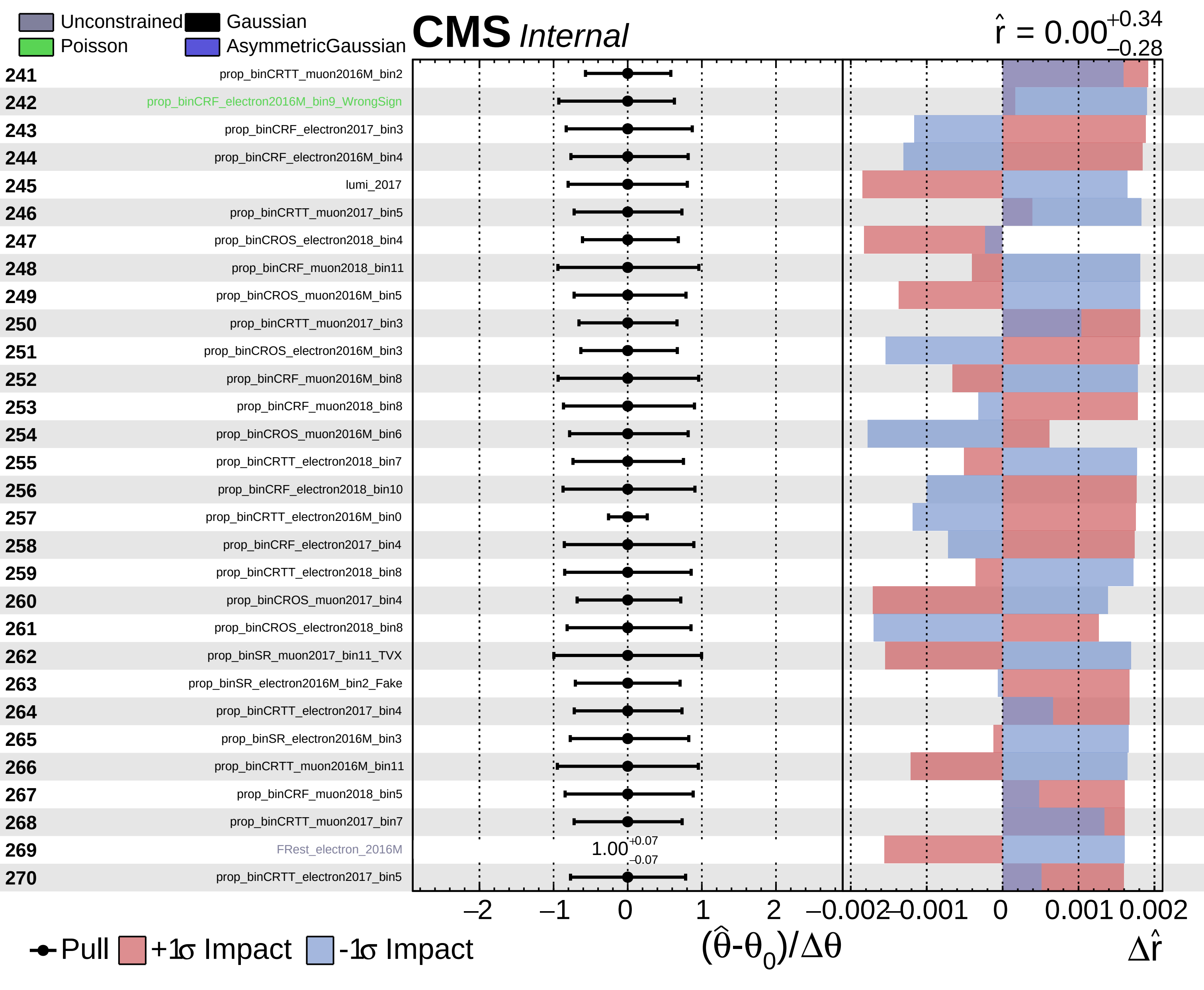
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.28
151 prop_binCRTT_muon2018_bin5
152 prop_binSR_electron2016M_bin5
153 prop_binCROS_electron2017_bin7_WrongSign
154 jer_2016M
155 prop_binSR_electron2016M_bin7
156 TTest_muon_2017
1.00 +0.05 −0.05
157 prop_binCRTT_muon2017_bin4
158 prop_binSR_muon2017_bin7_Fake
159 prop_binCRF_electron2017_bin10_TVX
160 prop_binCROS_electron2018_bin12_WrongSign
161 prop_binCRTT_muon2018_bin4
162 prop_binCRF_electron2016M_bin2
163 prop_binCROS_muon2018_bin7
164 QCDScale_TVX
165 prop_binSR_muon2017_bin7_WrongSign
166 prop_binCRTT_muon2018_bin1
167 prop_binCRTT_muon2018_bin9
168 prop_binSR_muon2016M_bin3
169 prop_binCRTT_electron2018_bin0
170 FRest_muon_2018
1.00 +0.06 −0.06
171 prop_binCRTT_muon2018_bin10
172 prop_binCROS_electron2017_bin3
173 prop_binCRTT_electron2018_bin1
174 prop_binCRF_muon2018_bin9
175 FRest_muon_2016M
1.00 +0.07 −0.07
176 prop_binSR_muon2018_bin3
177 prop_binCRTT_muon2018_bin11
178 prop_binCRF_muon2018_bin7
179 TTest_muon_2016M
1.00 +0.06 −0.06
180 prop_binCRF_muon2017_bin7
−2 −1 0 1 2
−0.004 −0.002 0 0.002 0.004
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

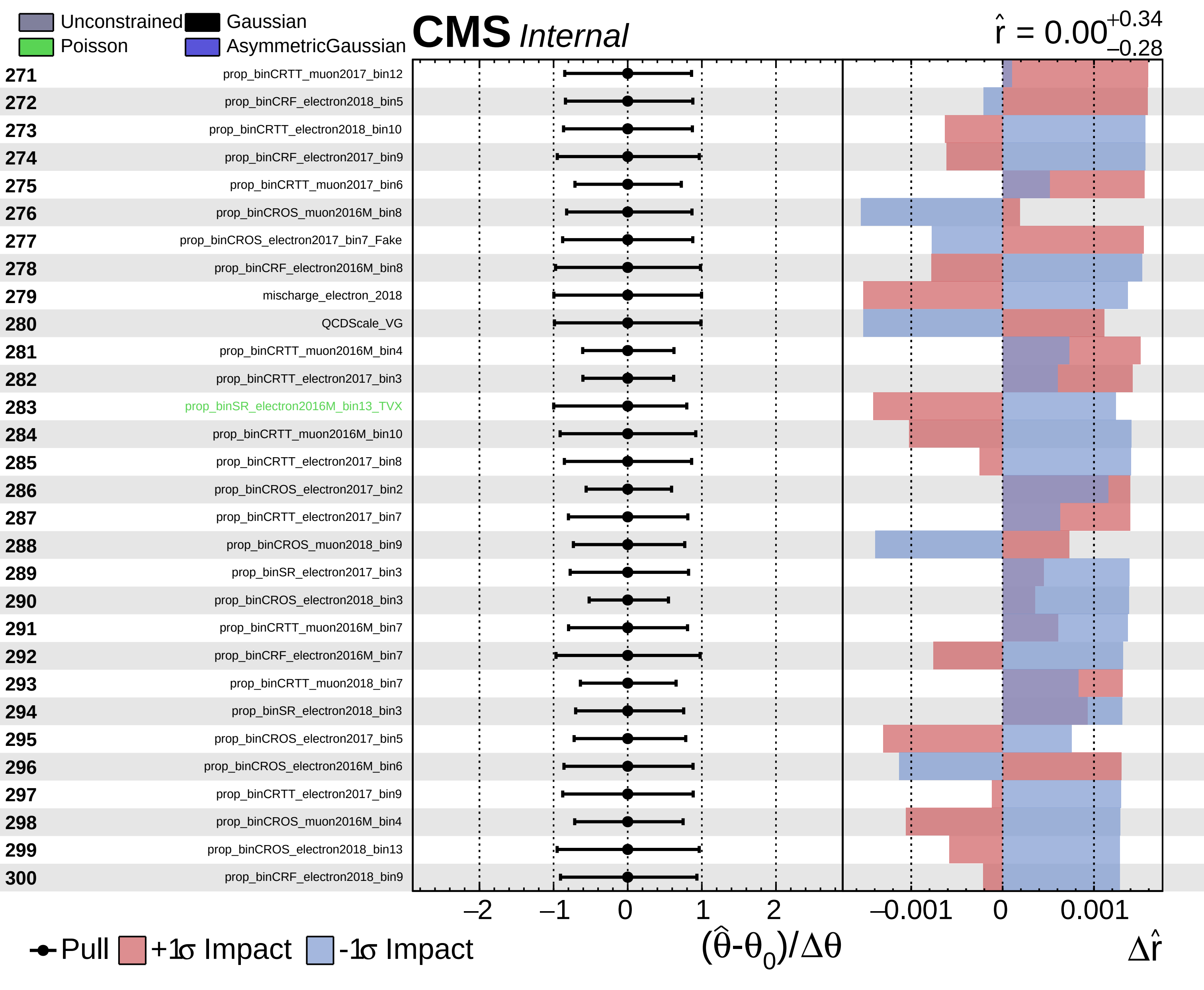


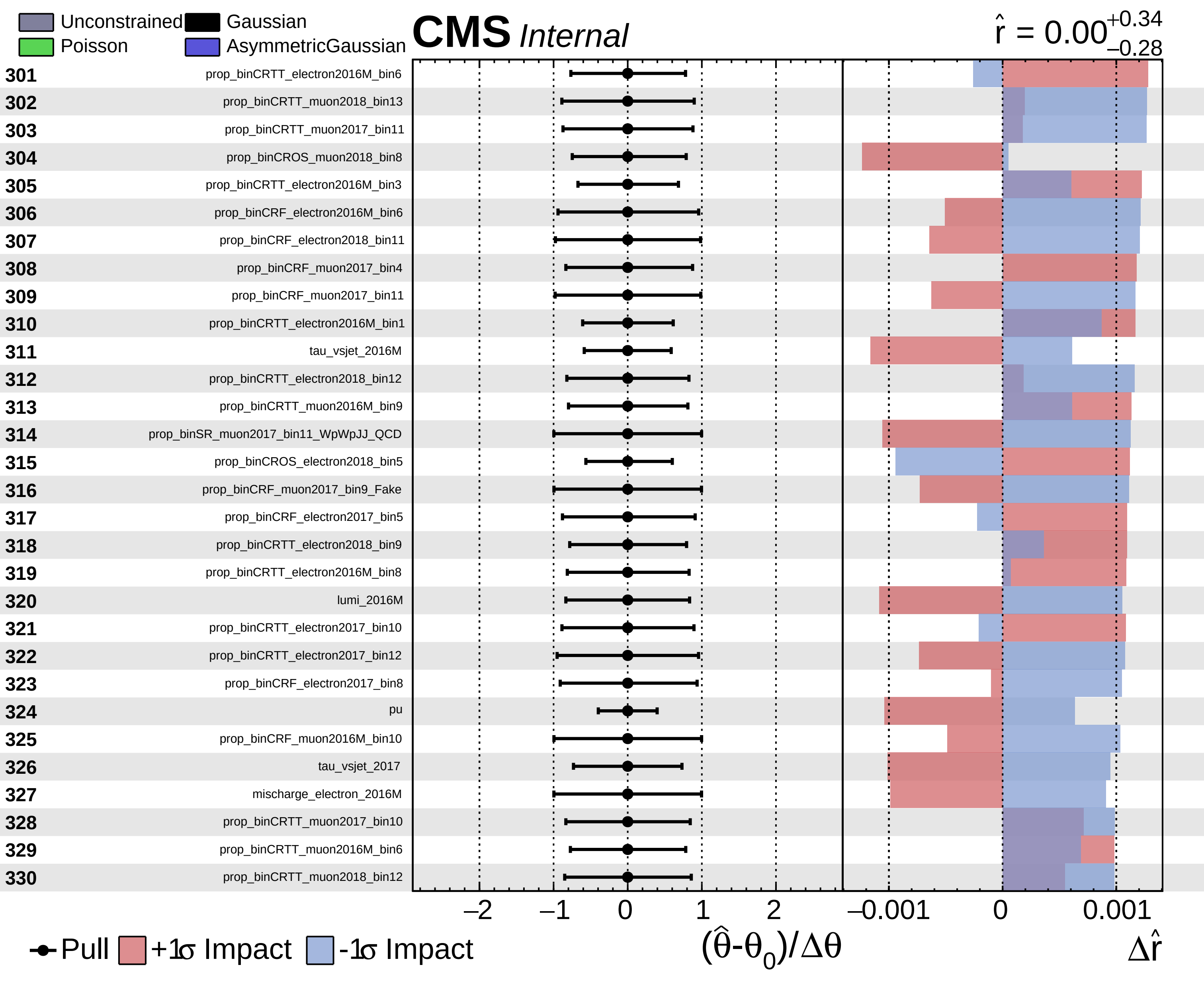
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
181 prop_binCRTT_muon2016M_bin3
182 QCDScale_Triboson
183 prop_binCRF_muon2017_bin10_Fake
184 mistag
185 prop_binCRTT_muon2016M_bin8
186 prop_binCROS_muon2017_bin1
187 prop_binSR_electron2016M_bin6
188 prop_binCROS_muon2017_bin8_WrongSign
189 mischarge_muon_2016M
190 prop_binSR_electron2018_bin4
191 prop_binCROS_electron2018_bin11_WrongSign
192 prop_binCROS_electron2016M_bin12_WrongSign
193 prop_binCRTT_muon2017_bin13
194 FRest_muon_2017
$1.00^{+0.07}_{-0.07}$
195 prop_binCRTT_electron2017_bin2
196 prop_binCRTT_muon2018_bin3
197 prop_binCRF_muon2016M_bin4
198 prop_binSR_electron2016M_bin1
199 jes_2016M
200 prop_binCRTT_electron2018_bin5
201 prop_binCROS_muon2018_bin10
202 prop_binSR_muon2017_bin13_TVX
203 FRest_electron_2018
$1.00^{+0.06}_{-0.06}$
204 prop_binCRTT_electron2016M_bin4
205 prop_binSR_muon2018_bin4
206 prop_binCROS_electron2016M_bin1
207 prop_binCROS_muon2018_bin6
208 FRest_electron_2017
$1.00^{+0.08}_{-0.07}$
209 prop_binCRTT_electron2018_bin6
210 prop_binCRTT_muon2016M_bin5
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



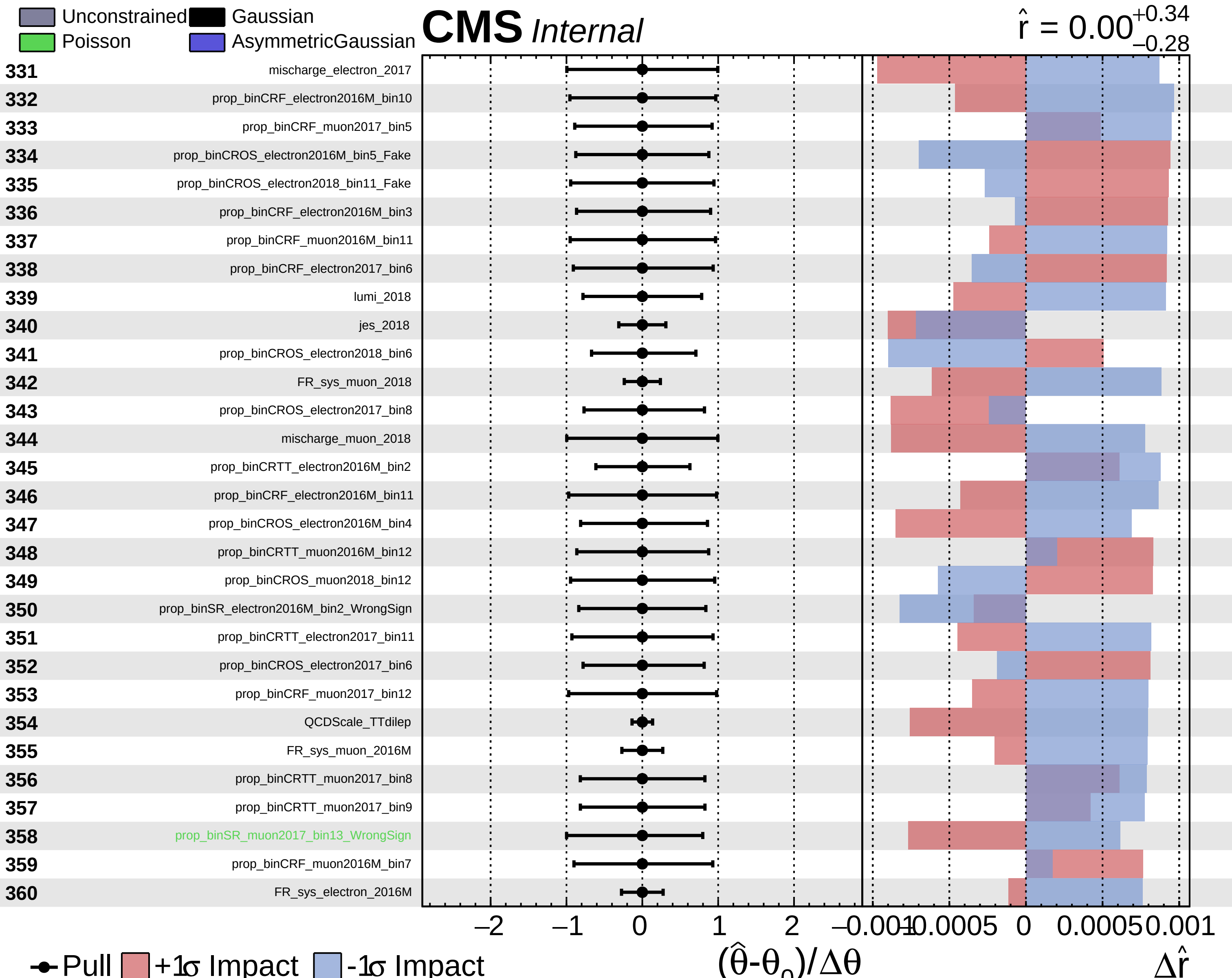
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
211 prop_binCRF_muon2017_bin3
212 prop_binCROS_muon2017_bin2
213 TTest_electron_2016M
$1.00^{+0.07}_{-0.07}$
214 prop_binCRTT_muon2018_bin8
215 prop_binCROS_electron2018_bin9
216 prop_binCRF_muon2017_bin6
217 prop_binCRTT_electron2017_bin6
218 prop_binCROS_electron2017_bin10
219 prop_binCRF_muon2016M_bin5
220 prop_binCRTT_electron2018_bin4
221 prop_binCRF_electron2018_bin8
222 prop_binSR_muon2017_bin13_WpWpJJ_QCD
223 prop_binCROS_electron2016M_bin8
224 prop_binCRF_electron2018_bin7
225 prop_binSR_electron2017_bin2
226 puID
227 prop_binCRF_electron2017_bin7
228 prop_binCROS_muon2018_bin2
229 TES
230 prop_binCRF_muon2018_bin10
231 prop_binCRTT_muon2017_bin1
232 prop_binSR_electron2018_bin2
233 prop_binCRF_muon2017_bin8
234 prop_binCROS_electron2018_bin1
235 prop_binCRF_muon2016M_bin9_ZZtoLep
236 prop_binCROS_muon2018_bin11_WrongSign
237 prop_binCRF_electron2018_bin6
238 prop_binCRTT_muon2018_bin6
239 prop_binCROS_muon2016M_bin7_WrongSign
240 prop_binSR_muon2017_bin13_Triboson
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

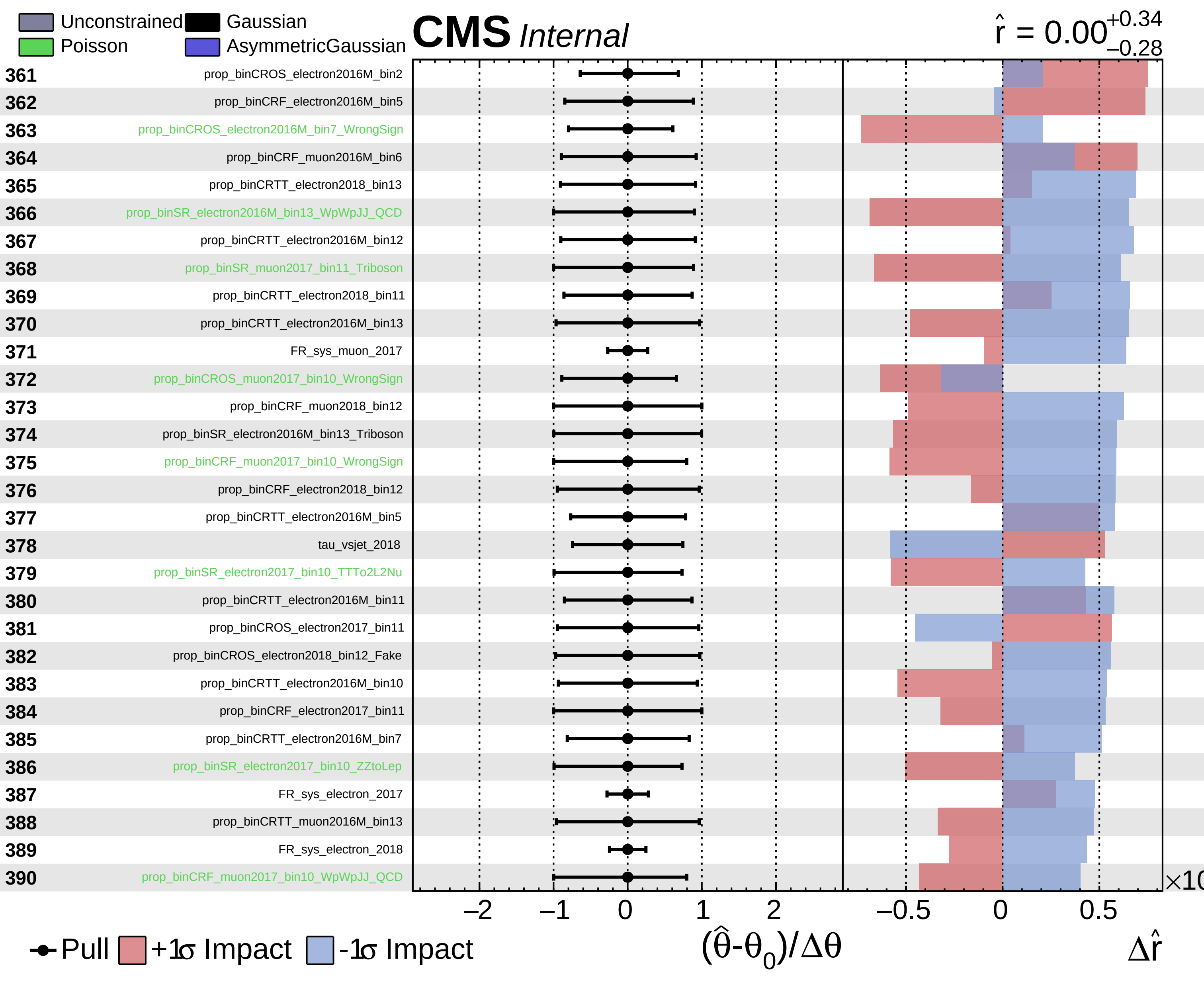

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
241 prop_binCRTT_muon2016M_bin2
242 prop_binCRF_electron2016M_bin9_WrongSign
243 prop_binCRF_electron2017_bin3
244 prop_binCRF_electron2016M_bin4
245 lumi_2017
246 prop_binCRTT_muon2017_bin5
247 prop_binCROS_electron2018_bin4
248 prop_binCRF_muon2018_bin11
249 prop_binCROS_muon2016M_bin5
250 prop_binCRTT_muon2017_bin3
251 prop_binCROS_electron2016M_bin3
252 prop_binCRF_muon2016M_bin8
253 prop_binCRF_muon2018_bin8
254 prop_binCROS_muon2016M_bin6
255 prop_binCRTT_electron2018_bin7
256 prop_binCRF_electron2018_bin10
257 prop_binCRTT_electron2016M_bin0
258 prop_binCRF_electron2017_bin4
259 prop_binCRTT_electron2018_bin8
260 prop_binCROS_muon2017_bin4
261 prop_binCROS_electron2018_bin8
262 prop_binSR_muon2017_bin11_TVX
263 prop_binSR_electron2016M_bin2_Fake
264 prop_binCRTT_electron2017_bin4
265 prop_binSR_electron2016M_bin3
266 prop_binCRTT_muon2016M_bin11
267 prop_binCRF_muon2018_bin5
268 prop_binCRTT_muon2017_bin7
269 FRest_electron_2016M
$1.00^{+0.07}_{-0.07}$
270 prop_binCRTT_electron2017_bin5
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
271 prop_binCRTT_muon2017_bin12
272 prop_binCRF_electron2018_bin5
273 prop_binCRTT_electron2018_bin10
274 prop_binCRF_electron2017_bin9
275 prop_binCRTT_muon2017_bin6
276 prop_binCROS_muon2016M_bin8
277 prop_binCROS_electron2017_bin7_Fake
278 prop_binCRF_electron2016M_bin8
279 mischarge_electron_2018
280 QCDScale_VG
281 prop_binCRTT_muon2016M_bin4
282 prop_binCRTT_electron2017_bin3
283 prop_binSR_electron2016M_bin13_TVX
284 prop_binCRTT_muon2016M_bin10
285 prop_binCRTT_electron2017_bin8
286 prop_binCROS_electron2017_bin2
287 prop_binCRTT_electron2017_bin7
288 prop_binCROS_muon2018_bin9
289 prop_binSR_electron2017_bin3
290 prop_binCROS_electron2018_bin3
291 prop_binCRTT_muon2016M_bin7
292 prop_binCRF_electron2016M_bin7
293 prop_binCRTT_muon2018_bin7
294 prop_binSR_electron2018_bin3
295 prop_binCROS_electron2017_bin5
296 prop_binCROS_electron2016M_bin6
297 prop_binCRTT_electron2017_bin9
298 prop_binCROS_muon2016M_bin4
299 prop_binCROS_electron2018_bin13
300 prop_binCRF_electron2018_bin9
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

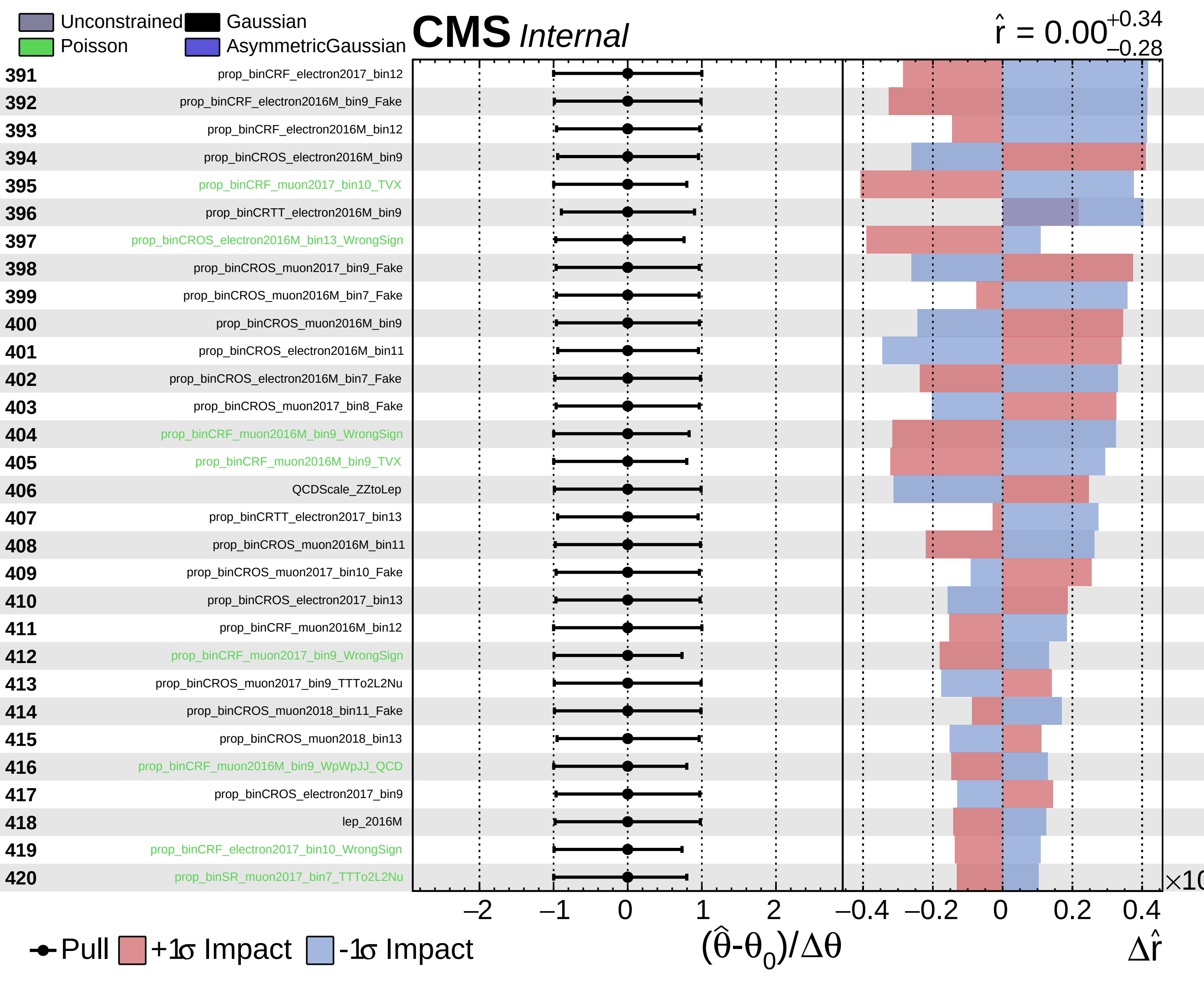


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
301 prop_binCRTT_electron2016M_bin6
302 prop_binCRTT_muon2018_bin13
303 prop_binCRTT_muon2017_bin11
304 prop_binCROS_muon2018_bin8
305 prop_binCRTT_electron2016M_bin3
306 prop_binCRF_electron2016M_bin6
307 prop_binCRF_electron2018_bin11
308 prop_binCRF_muon2017_bin4
309 prop_binCRF_muon2017_bin11
310 prop_binCRTT_electron2016M_bin1
311 tau_vsjet_2016M
312 prop_binCRTT_electron2018_bin12
313 prop_binCRTT_muon2016M_bin9
314 prop_binSR_muon2017_bin11_WpWpJJ_QCD
315 prop_binCROS_electron2018_bin5
316 prop_binCRF_muon2017_bin9_Fake
317 prop_binCRF_electron2017_bin5
318 prop_binCRTT_electron2018_bin9
319 prop_binCRTT_electron2016M_bin8
320 lumi_2016M
321 prop_binCRTT_electron2017_bin10
322 prop_binCRTT_electron2017_bin12
323 prop_binCRF_electron2017_bin8
324 pu
325 prop_binCRF_muon2016M_bin10
326 tau_vsjet_2017
327 mischarge_electron_2016M
328 prop_binCRTT_muon2017_bin10
329 prop_binCRTT_muon2016M_bin6
330 prop_binCRTT_muon2018_bin12
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

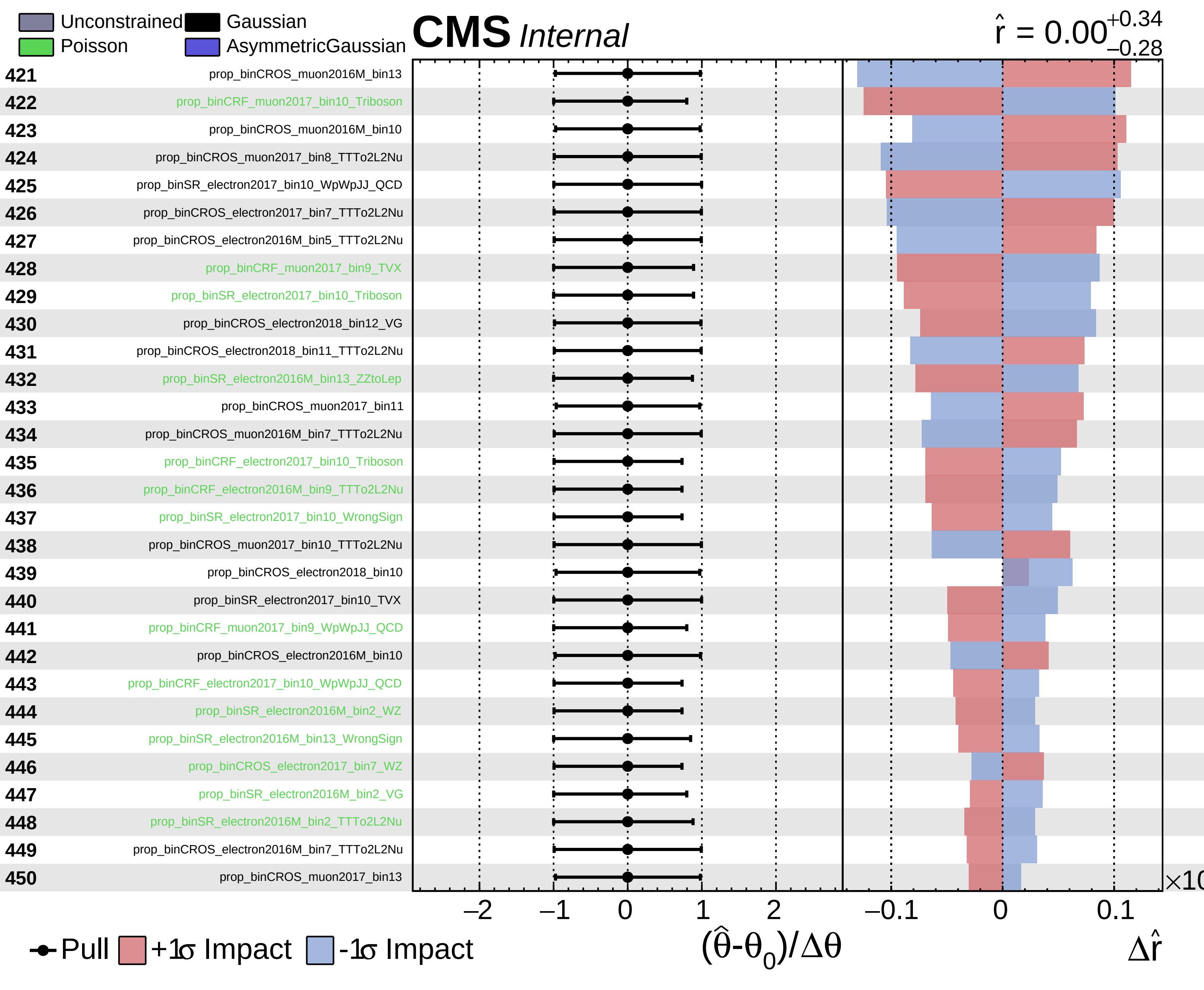

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
331 mischarge_electron_2017
332 prop_binCRF_electron2016M_bin10
333 prop_binCRF_muon2017_bin5
334 prop_binCROS_electron2016M_bin5_Fake
335 prop_binCROS_electron2018_bin11_Fake
336 prop_binCRF_electron2016M_bin3
337 prop_binCRF_muon2016M_bin11
338 prop_binCRF_electron2017_bin6
339 lumi_2018
340 jes_2018
341 prop_binCROS_electron2018_bin6
342 FR_sys_muon_2018
343 prop_binCROS_electron2017_bin8
344 mischarge_muon_2018
345 prop_binCRTT_electron2016M_bin2
346 prop_binCRF_electron2016M_bin11
347 prop_binCROS_electron2016M_bin4
348 prop_binCRTT_muon2016M_bin12
349 prop_binCROS_muon2018_bin12
350 prop_binSR_electron2016M_bin2_WrongSign
351 prop_binCRTT_electron2017_bin11
352 prop_binCROS_electron2017_bin6
353 prop_binCRF_muon2017_bin12
354 QCDScale_TTdilep
355 FR_sys_muon_2016M
356 prop_binCRTT_muon2017_bin8
357 prop_binCRTT_muon2017_bin9
358 prop_binSR_muon2017_bin13_WrongSign
359 prop_binCRF_muon2016M_bin7
360 FR_sys_electron_2016M
−2 −1 0 1 2
−0.001 −0.0005 0 0.0005 0.001
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

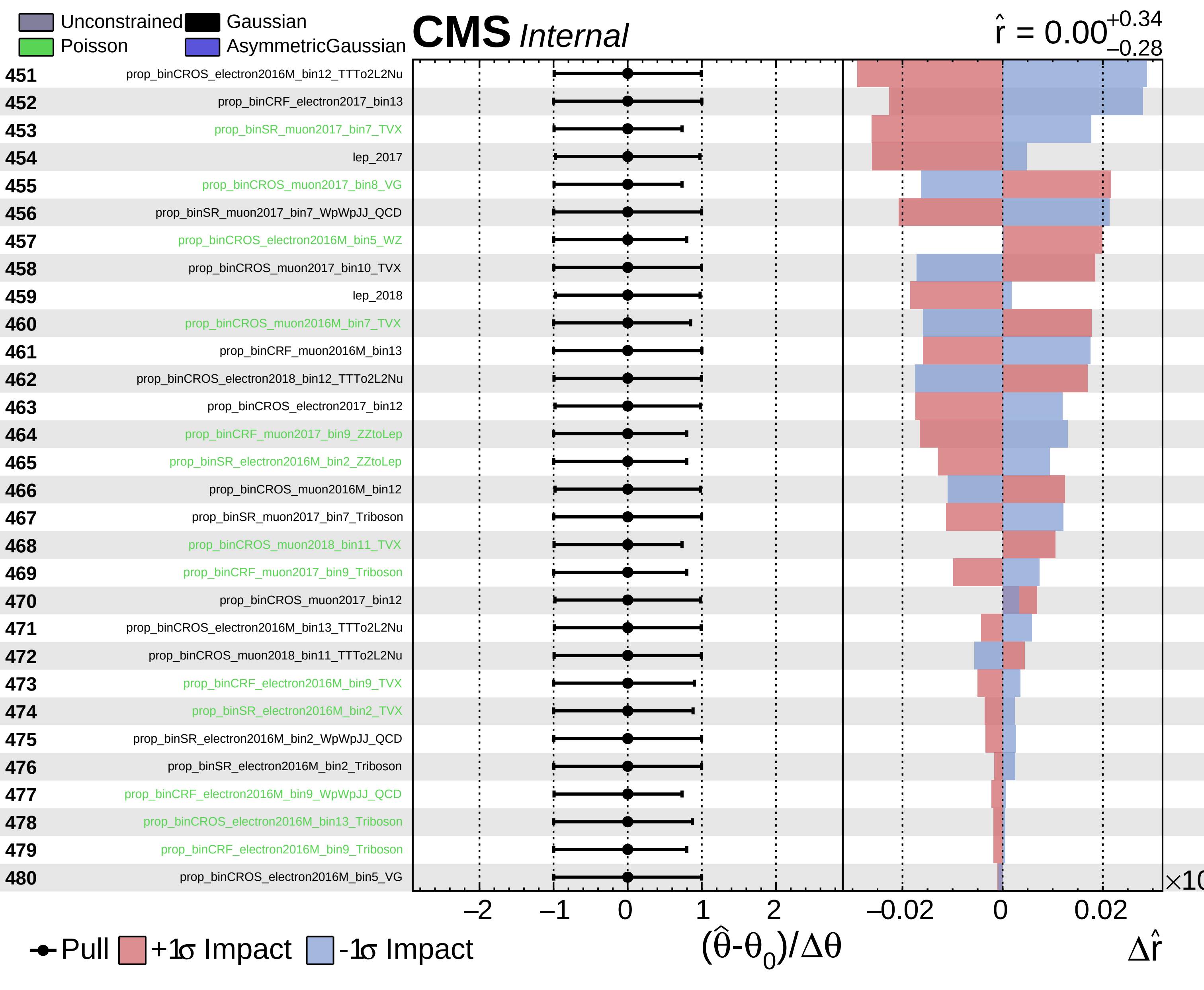


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
361 prop_binCROS_electron2016M_bin2
362 prop_binCRF_electron2016M_bin5
363 prop_binCROS_electron2016M_bin7_WrongSign
364 prop_binCRF_muon2016M_bin6
365 prop_binCRTT_electron2018_bin13
366 prop_binSR_electron2016M_bin13_WpWpJJ_QCD
367 prop_binCRTT_electron2016M_bin12
368 prop_binSR_muon2017_bin11_Triboson
369 prop_binCRTT_electron2018_bin11
370 prop_binCRTT_electron2016M_bin13
371 FR_sys_muon_2017
372 prop_binCROS_muon2017_bin10_WrongSign
373 prop_binCRF_muon2018_bin12
374 prop_binSR_electron2016M_bin13_Triboson
375 prop_binCRF_muon2017_bin10_WrongSign
376 prop_binCRF_electron2018_bin12
377 prop_binCRTT_electron2016M_bin5
378 tau_vsjet_2018
379 prop_binSR_electron2017_bin10_TTTo2L2Nu
380 prop_binCRTT_electron2016M_bin11
381 prop_binCROS_electron2017_bin11
382 prop_binCROS_electron2018_bin12_Fake
383 prop_binCRTT_electron2016M_bin10
384 prop_binCRF_electron2017_bin11
385 prop_binCRTT_electron2016M_bin7
386 prop_binSR_electron2017_bin10_ZZtoLep
387 FR_sys_electron_2017
388 prop_binCRTT_muon2016M_bin13
389 FR_sys_electron_2018
390 prop_binCRF_muon2017_bin10_WpWpJJ_QCD
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

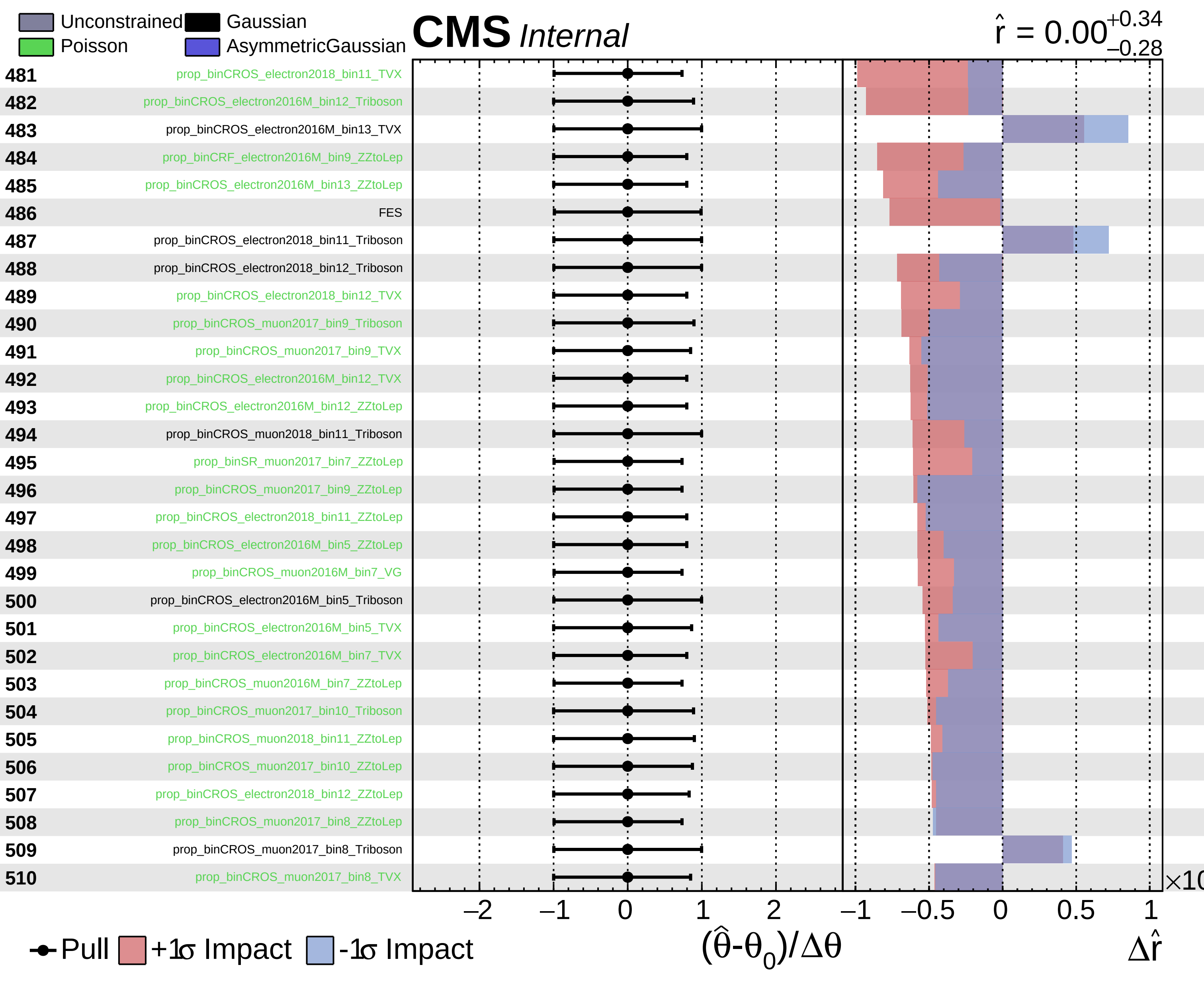


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = 0.00 +0.34 −0.28
391 prop_binCRF_electron2017_bin12
392 prop_binCRF_electron2016M_bin9_Fake
393 prop_binCRF_electron2016M_bin12
394 prop_binCROS_electron2016M_bin9
395 prop_binCRF_muon2017_bin10_TVX
396 prop_binCRTT_electron2016M_bin9
397 prop_binCROS_electron2016M_bin13_WrongSign
398 prop_binCROS_muon2017_bin9_Fake
399 prop_binCROS_muon2016M_bin7_Fake
400 prop_binCROS_muon2016M_bin9
401 prop_binCROS_electron2016M_bin11
402 prop_binCROS_electron2016M_bin7_Fake
403 prop_binCROS_muon2017_bin8_Fake
404 prop_binCRF_muon2016M_bin9_WrongSign
405 prop_binCRF_muon2016M_bin9_TVX
406 QCDScale_ZZtoLep
407 prop_binCRTT_electron2017_bin13
408 prop_binCROS_muon2016M_bin11
409 prop_binCROS_muon2017_bin10_Fake
410 prop_binCROS_electron2017_bin13
411 prop_binCRF_muon2016M_bin12
412 prop_binCRF_muon2017_bin9_WrongSign
413 prop_binCROS_muon2017_bin9_TTTo2L2Nu
414 prop_binCROS_muon2018_bin11_Fake
415 prop_binCROS_muon2018_bin13
416 prop_binCRF_muon2016M_bin9_WpWpJJ_QCD
417 prop_binCROS_electron2017_bin9
418 lep_2016M
419 prop_binCRF_electron2017_bin10_WrongSign
420 prop_binSR_muon2017_bin7_TTTo2L2Nu
×10
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

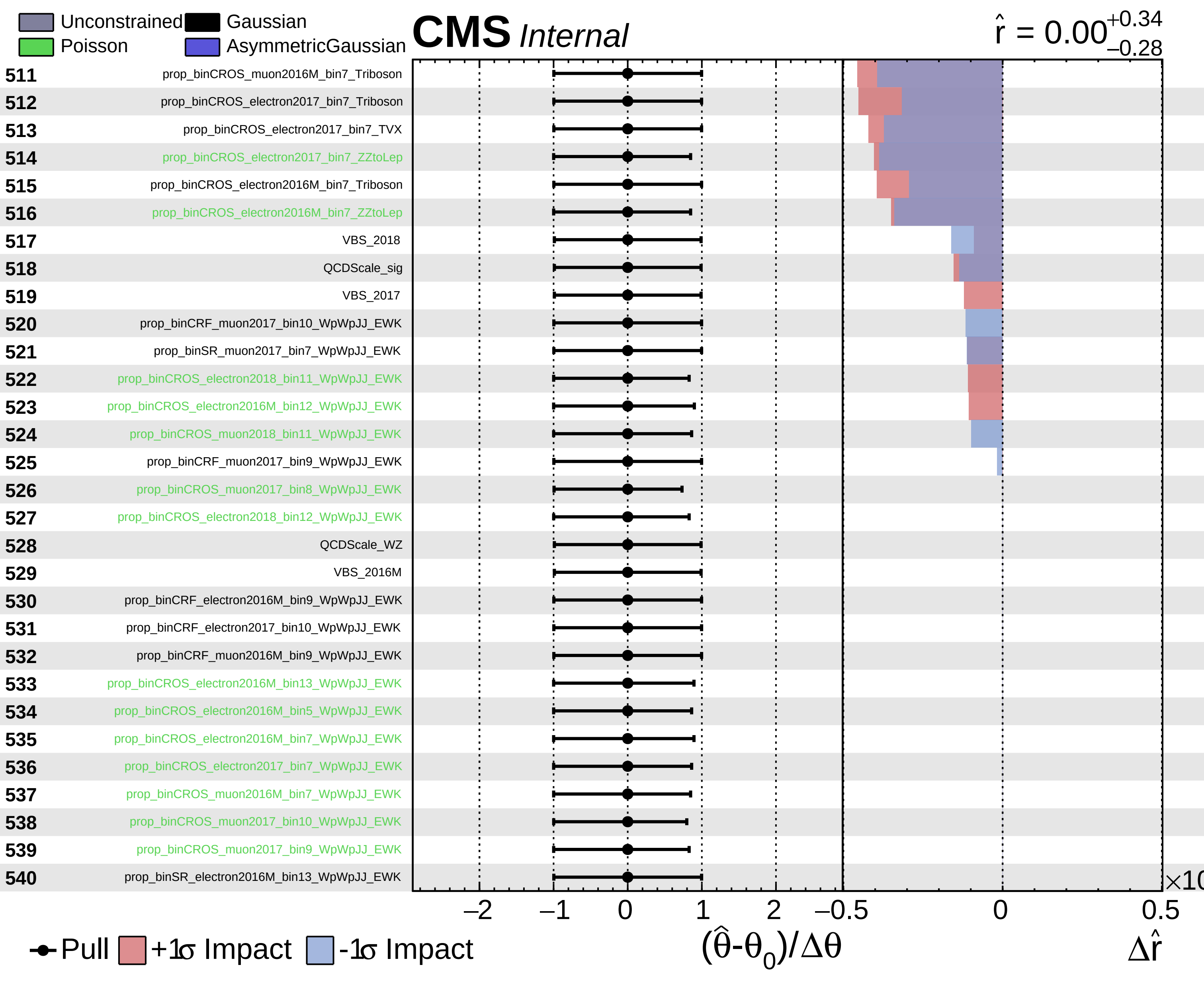

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
421 prop_binCROS_muon2016M_bin13
422 prop_binCRF_muon2017_bin10_Triboson
423 prop_binCROS_muon2016M_bin10
424 prop_binCROS_muon2017_bin8_TTTo2L2Nu
425 prop_binSR_electron2017_bin10_WpWpJJ_QCD
426 prop_binCROS_electron2017_bin7_TTTo2L2Nu
427 prop_binCROS_electron2016M_bin5_TTTo2L2Nu
428 prop_binCRF_muon2017_bin9_TVX
429 prop_binSR_electron2017_bin10_Triboson
430 prop_binCROS_electron2018_bin12_VG
431 prop_binCROS_electron2018_bin11_TTTo2L2Nu
432 prop_binSR_electron2016M_bin13_ZZtoLep
433 prop_binCROS_muon2017_bin11
434 prop_binCROS_muon2016M_bin7_TTTo2L2Nu
435 prop_binCRF_electron2017_bin10_Triboson
436 prop_binCRF_electron2016M_bin9_TTTo2L2Nu
437 prop_binSR_electron2017_bin10_WrongSign
438 prop_binCROS_muon2017_bin10_TTTo2L2Nu
439 prop_binCROS_electron2018_bin10
440 prop_binSR_electron2017_bin10_TVX
441 prop_binCRF_muon2017_bin9_WpWpJJ_QCD
442 prop_binCROS_electron2016M_bin10
443 prop_binCRF_electron2017_bin10_WpWpJJ_QCD
444 prop_binSR_electron2016M_bin2_WZ
445 prop_binSR_electron2016M_bin13_WrongSign
446 prop_binCROS_electron2017_bin7_WZ
447 prop_binSR_electron2016M_bin2_VG
448 prop_binSR_electron2016M_bin2_TTTo2L2Nu
449 prop_binCROS_electron2016M_bin7_TTTo2L2Nu
450 prop_binCROS_muon2017_bin13
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



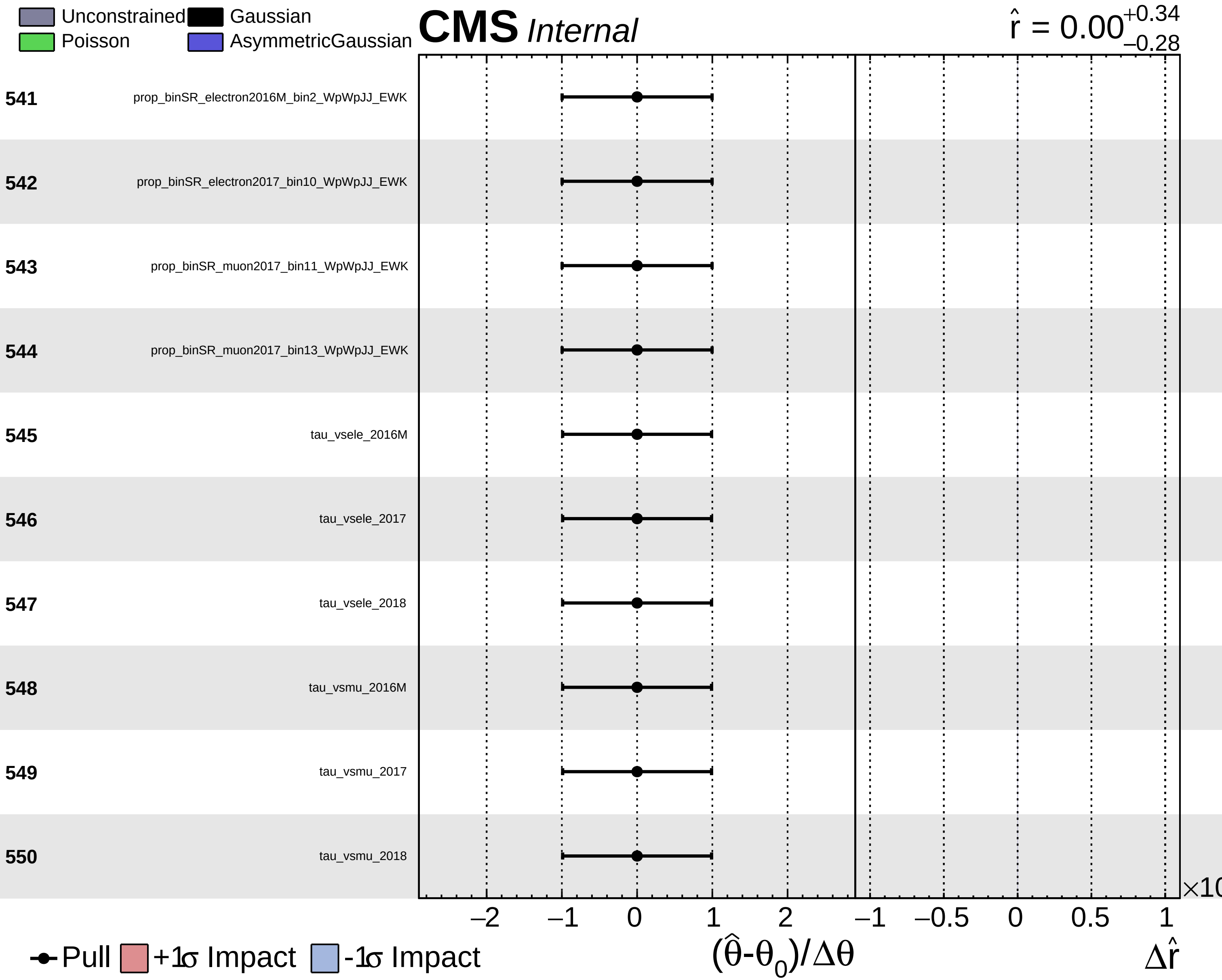
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
451 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
452 prop_binCRF_electron2017_bin13
453 prop_binSR_muon2017_bin7_TVX
454 lep_2017
455 prop_binCROS_muon2017_bin8_VG
456 prop_binSR_muon2017_bin7_WpWpJJ_QCD
457 prop_binCROS_electron2016M_bin5_WZ
458 prop_binCROS_muon2017_bin10_TVX
459 lep_2018
460 prop_binCROS_muon2016M_bin7_TVX
461 prop_binCRF_muon2016M_bin13
462 prop_binCROS_electron2018_bin12_TTTo2L2Nu
463 prop_binCROS_electron2017_bin12
464 prop_binCRF_muon2017_bin9_ZZtoLep
465 prop_binSR_electron2016M_bin2_ZZtoLep
466 prop_binCROS_muon2016M_bin12
467 prop_binSR_muon2017_bin7_Triboson
468 prop_binCROS_muon2018_bin11_TVX
469 prop_binCRF_muon2017_bin9_Triboson
470 prop_binCROS_muon2017_bin12
471 prop_binCROS_electron2016M_bin13_TTTo2L2Nu
472 prop_binCROS_muon2018_bin11_TTTo2L2Nu
473 prop_binCRF_electron2016M_bin9_TVX
474 prop_binSR_electron2016M_bin2_TVX
475 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
476 prop_binSR_electron2016M_bin2_Triboson
477 prop_binCRF_electron2016M_bin9_WpWpJJ_QCD
478 prop_binCROS_electron2016M_bin13_Triboson
479 prop_binCRF_electron2016M_bin9_Triboson
480 prop_binCROS_electron2016M_bin5_VG
−2 −1 0 1 2
−0.02 0 0.02
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
481 prop_binCROS_electron2018_bin11_TVX
482 prop_binCROS_electron2016M_bin12_Triboson
483 prop_binCROS_electron2016M_bin13_TVX
484 prop_binCRF_electron2016M_bin9_ZZtoLep
485 prop_binCROS_electron2016M_bin13_ZZtoLep
486 FES
487 prop_binCROS_electron2018_bin11_Triboson
488 prop_binCROS_electron2018_bin12_Triboson
489 prop_binCROS_electron2018_bin12_TVX
490 prop_binCROS_muon2017_bin9_Triboson
491 prop_binCROS_muon2017_bin9_TVX
492 prop_binCROS_electron2016M_bin12_TVX
493 prop_binCROS_electron2016M_bin12_ZZtoLep
494 prop_binCROS_muon2018_bin11_Triboson
495 prop_binSR_muon2017_bin7_ZZtoLep
496 prop_binCROS_muon2017_bin9_ZZtoLep
497 prop_binCROS_electron2018_bin11_ZZtoLep
498 prop_binCROS_electron2016M_bin5_ZZtoLep
499 prop_binCROS_muon2016M_bin7_VG
500 prop_binCROS_electron2016M_bin5_Triboson
501 prop_binCROS_electron2016M_bin5_TVX
502 prop_binCROS_electron2016M_bin7_TVX
503 prop_binCROS_muon2016M_bin7_ZZtoLep
504 prop_binCROS_muon2017_bin10_Triboson
505 prop_binCROS_muon2018_bin11_ZZtoLep
506 prop_binCROS_muon2017_bin10_ZZtoLep
507 prop_binCROS_electron2018_bin12_ZZtoLep
508 prop_binCROS_muon2017_bin8_ZZtoLep
509 prop_binCROS_muon2017_bin8_Triboson
510 prop_binCROS_muon2017_bin8_TVX
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
511 prop_binCROS_muon2016M_bin7_Triboson
512 prop_binCROS_electron2017_bin7_Triboson
513 prop_binCROS_electron2017_bin7_TVX
514 prop_binCROS_electron2017_bin7_ZZtoLep
515 prop_binCROS_electron2016M_bin7_Triboson
516 prop_binCROS_electron2016M_bin7_ZZtoLep
517 VBS_2018
518 QCDScale_sig
519 VBS_2017
520 prop_binCRF_muon2017_bin10_WpWpJJ_EWK
521 prop_binSR_muon2017_bin7_WpWpJJ_EWK
522 prop_binCROS_electron2018_bin11_WpWpJJ_EWK
523 prop_binCROS_electron2016M_bin12_WpWpJJ_EWK
524 prop_binCROS_muon2018_bin11_WpWpJJ_EWK
525 prop_binCRF_muon2017_bin9_WpWpJJ_EWK
526 prop_binCROS_muon2017_bin8_WpWpJJ_EWK
527 prop_binCROS_electron2018_bin12_WpWpJJ_EWK
528 QCDScale_WZ
529 VBS_2016M
530 prop_binCRF_electron2016M_bin9_WpWpJJ_EWK
531 prop_binCRF_electron2017_bin10_WpWpJJ_EWK
532 prop_binCRF_muon2016M_bin9_WpWpJJ_EWK
533 prop_binCROS_electron2016M_bin13_WpWpJJ_EWK
534 prop_binCROS_electron2016M_bin5_WpWpJJ_EWK
535 prop_binCROS_electron2016M_bin7_WpWpJJ_EWK
536 prop_binCROS_electron2017_bin7_WpWpJJ_EWK
537 prop_binCROS_muon2016M_bin7_WpWpJJ_EWK
538 prop_binCROS_muon2017_bin10_WpWpJJ_EWK
539 prop_binCROS_muon2017_bin9_WpWpJJ_EWK
540 prop_binSR_electron2016M_bin13_WpWpJJ_EWK
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = 0.00^{+0.34}_{-0.28}$
541 prop_binSR_electron2016M_bin2_WpWpJJ_EWK
542 prop_binSR_electron2017_bin10_WpWpJJ_EWK
543 prop_binSR_muon2017_bin11_WpWpJJ_EWK
544 prop_binSR_muon2017_bin13_WpWpJJ_EWK
545 tau_vsele_2016M
546 tau_vsele_2017
547 tau_vsele_2018
548 tau_vsmu_2016M
549 tau_vsmu_2017
550 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$